

Department of Computer Science and Business Systems

Academic Year 2024– 2025 (Even Semester)

Degree, Semester & Branch: VI Semester – B.Tech CSBS

Course Code & Title: CCS356- Object Oriented Software Engineering

Name of the Faculty member (s): Ms.M.Preethi Ram, AP/CSBS

Innovative Practice Description

- **Unit / Topic: Unit II / Formal System Specification**
- **Course Outcome: CO2**
- **Topic Learning Outcome /UO: 2a**
- **Activity Chosen: One Minute Paper**
- **Justification:**
 - In one-minute paper activity the students will be given with 5 minutes to write the brief about that class session.
 - A successful Content marketing strategy typically consists of several key modules that work together to achieve specific goals.
 - In order to have a recap of content marketing strategy modules this activity is chosen.
- **Time Allotted for the Activity: 15 Minutes**
- **Details of the Implementation:**
 - Faculty explained the concept of Formal system Specification
 - Students were asked to summarize the important contents in formal system specification in a paper within a minute as shown in fig 1.
 - The faculty member collected and corrected the answer scripts as shown in fig 2.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PSO1	PSO2
CO2	2	2	1	1	1	2

(1 – Low 2 – Moderate 3 – High)



• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PSO1	PSO2
Justification for correlation	Students apply mathematical models and logical reasoning to specify complex software systems formally.	Students identify inconsistencies and ambiguities in requirements and specify precise system behavior.	Students design robust and verifiable software systems using formal methods like Z notation or FSM.	Students verify software specifications, simulate use cases, and validate through formal proofs or tools.	Formal specifications help build accurate models that support AI decision logic or cloud-based software systems.	Formal specifications are integral in model-driven development and ensure error-free design logic in software systems.

• **Images / Screenshot of the practice:**

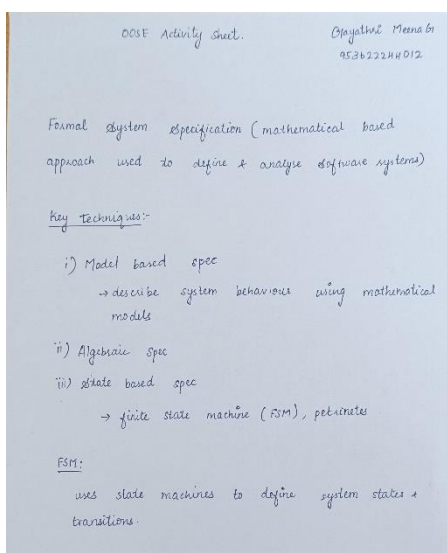


Fig 1: Sample Sheet of activity of the Student- Gayathri Meena



Fig 2: Students photo while they are recollecting the concept contents



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- ❖ Students said that they were able to recollect and remember the concepts of formal system specification after the activity.

- ❖ Benefit of the practice:

- Students' remembrance of the concept is improved
 - Get a sense of what the students have learned, where there might be gaps in their knowledge

- ❖ ***Challenges faced in implementation:***

- Few students made a mistake in writing the concept.
 - It took much time for some students to complete the activity

References:

- <https://www.slideshare.net/raja2dlas/minute-paper>
- <https://www.ritrjpm.ac.in/images/computer-science/CS8251-SE-MinutePaper.pdf>